DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (GAI) IN THE MASTER'S THESIS (TFM)

I have used GAI in my TFM
Check the appropriate box:

YES	NO

If you checked YES, please complete the following 3 sections of this document:

Part 1: Declaration of Legal, Ethical, and Responsible Conduct

Please be aware that the use of GAI tools entails certain risks and may give rise to serious academic consequences. The evaluation of your TFM may be compromised if GAI use involves confidential data, copyright-protected materials, or personal data handled without fulfilling the required conditions in each case (i.e., without the consent of the data subjects, the authorization of rights holders, or in breach of the University's guidelines).

<table>
<tr><th colspan="2">Question</th></tr>
<tr><td colspan="2">• In my interactions with GAI tools, I have provided confidential data with the appropriate authorization from the parties concerned. Confidential data encompasses any information that an individual or organization seeks to protect for legal, commercial, privacy, or strategic reasons (such as patents or trade secrets).</td></tr>
<tr><td>YES, I used this data with the consent of the parties concerned</td><td>NO, I did not use any confidential data</td></tr>
<tr><td colspan="2">• In my interactions with GAI tools, I have provided copyright-protected materials with the authorization of the respective rights holders.</td></tr>
<tr><td>YES, I used these materials with authorization from the rights holders; or without it because they fall within one of the legal exceptions or limitations permitted by law:
• Works in the public domain
• Licensed works (Creative Commons licences)
• Use of excerpts for research purposes (right of quotation)</td><td>NO, I did not use any copyright-protected materials</td></tr>
<tr><td colspan="2">• In my interactions with GAI tools, I have provided personal data with the proper consent of the data subjects.</td></tr>
<tr><td>YES, I used this data with the consent of the data subjects and in accordance with the instructions set out in the University-approved guide</td><td>NO, I did not use any personal data</td></tr>
<tr><td colspan="2">• My use of GAI tools has complied with their terms of use and with fundamental ethical principles. I have not directed them in a malicious manner to obtain results inappropriate to the work presented, such as results that misrepresent the actual</td></tr>
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findings, substitute my own intellectual contribution, or could cause harm to individuals.	
YES	NO

Part 2: Technical Declaration of Use

Use the following declaration template as many times as necessary to account for all types of interaction you have had with GAI tools. Include one example per type of use in the spaces indicated: [Add an example].

I hereby declare that I have made use of the GAI system (Name of AI system/tool and version: ChatGPT, Gemini, Copilot...) for:

Documentation and Writing:

- *Support for reflection throughout the development of the work: iterative process of analysing alternatives and approaches using GAI*

Used as a brainstorming tool to analyse architectural design trade-offs. For example, discussions were held regarding the suitability of adopting a hybrid approach combining machine learning and LLM models versus relying solely on an LLM.

- *Review or rewriting of previously drafted paragraphs*

Since the thesis was written entirely in technical English, GAI was used to refine the grammar and elevate the academic tone of texts previously drafted by the author. For example, in "Chapter 5: Evaluation and Results", it was used to ensure that the description of metrics (such as the distinction between F1-Macro and F1-Weighted in imbalanced datasets) employed the precision and vocabulary expected of an IEEE-style article.

- *Information retrieval and responses to specific technical questions*

Resolution of specific technical queries regarding the advanced LaTeX syntax required for the IEEEtran double-column format.

- *Bibliography search*
- *Summarising consulted bibliography*
- *Translation of consulted texts*

Developing Specific Content

GAI was used as a support tool for developing the specific content of the TFM, including:

- *Assistance with code development (programming)*

Assistance in optimising the Python code, specifically in the management of threads (threading) and concurrency within the Orchestrator module.

- *Generation of diagrams, images, audio, or video*

Not applicable. No Generative AI tools were utilized to create or modify any figures, diagrams, or multimedia elements in this thesis.

- *Optimisation processes*

Optimisation and refactoring of the RAG (Retrieval-Augmented Generation) architecture code, implementing GAI-suggested improvements for exception handling and API retry logic, thereby enhancing the fault tolerance of the "Responder" agent.

- *Data handling: collection, analysis, cross-referencing of data, etc.*

Once the network traffic datasets (Edge-IIoTset, TON_IoT, and Bot-IoT) were gathered, the tool was consulted to obtain guidance on data normalization. Specifically, it assisted in designing the logic for cross-referencing and mapping heterogeneous network schemas into the 12 universal canonical features used in Phase 1.

- *Creative ideation and inspiration*

Assistance in structuring the “Future Work” section. The tool was consulted to explore contemporary extensions of the current work, which led to the inclusion of concepts such the evolution of the system towards an interactive Conversational SOC Assistant (Explainable AI).

Part 3: Reflection on Utility

Provide a personal assessment (free format) of the strengths and weaknesses you have identified in the use of GAI tools during the development of your work. Indicate whether they proved useful in the learning process, in the development of the work, or in drawing conclusions.

The use of Generative Artificial Intelligence tools as assistants throughout the development of this TFM allowed me to verify, in a practical setting, what these tools do well as academic support aids and where they fall short.

As an assistant, GAI saved me considerable time on debugging tasks. It proved particularly useful for resolving concurrency issues in the Python code and for configuring the double-column formatting of the LaTeX template. Removing those mechanical technical blockers freed me to focus my time and energy on what truly mattered: designing the architecture and analysing the experimental data. It also provided valuable support in refining the grammar and academic tone of the final document, written entirely in technical English.

On the other hand, using these assistants for programming clearly exposed their limitations. I found that when asked to generate complex code, they frequently produce generic solutions that do not work in a real-world environment without substantial modification. They also tend to hallucinate, inventing functions or disregarding the project context, when prompts are not formulated with sufficient precision. This demonstrates that GAI cannot replace the programmer's own expertise.

In summary, AI proved to be a valuable tool for working more efficiently, overcoming technical obstacles, and improving the formal presentation of the document. It is clear, however, that it serves only as a supplement: it accelerates the work, but the architectural design decisions, scientific rigour, data interpretation, and final validation that things actually function as intended depended exclusively on my own effort, critical oversight, and technical judgement.